Receive-Coherent Synthesis Method of Moving Target for Airborne Distributed Coherent Aperture Radar Based on Parameter Space Division

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Airborne distributed coherent aperture radar (ADCAR) benefits from the signal coherent synthesis of multichannel echoes to improve signal gain and detection performance significantly. However, the complex detection environment formed by the geometrical configuration between ADCAR and the region of interest always causes distance–velocity envelope shifts with different properties and relative phase deviations, severely reducing the signal coherent synthesis efficiency. We make contributions toward addressing these limitations. First, the signal model and description of target coherent parameters (TCPs) for ADCAR in the receive-coherent (RC) stage are considered, and a parameter space division using the dimensionality reduction (DR) condition for TCPs is proposed. Then, a multiple transmit–receive pairs joint TCPs estimation method based on maximum likelihood estimation is proposed, and the RC synthesis strategies are performed with the DR and non-DR conditions by the proposed TCPs estimation method. On this basis, two fast implementations based on particle swarm optimization and conjugate gradient are proposed for the RC synthesis and the TCPs output. Finally, the results of simulation experiments and real-measured data indicated that the performance of the proposed methods for signal coherent synthesis, TCPs estimation, and target detection is significantly improved compared with the traditional methods.

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NOMENCLATURE

ADCAR	Airborne distributed coherent aperture radar.
CAF	Cross ambiguity function.
CAN	Cyclic algorithms-new.
CC	Cross correlation.
CG	Conjugate gradient.
CPI	Coherent processing interval.
CRB	Cramer–Rao bound.
DCAR	Distributed coherent aperture radar.
DR	Dimensionality reduction.
FFT	Fast Fourier transform.
HCD	Hybrid coherent detector.
MF	Matcher filter.
MIMO	Multi-input–multi-output.
MLE	Maximum likelihood estimation.
NDR	Nondimensionality reduction.
PCM	Pulse-code modulation.
PP	Peak picking.
PRF	Pulse repetition frequency.
PSO	Particle swarm optimization.
RC	Receive coherent.
RMSE	Root mean square error.
SNR	Signal-to-noise ratio.
STAP	Space time adaptive processing.
TCPs	Target coherent parameters.
TR	Transmit–receive.
TRC	Transmit–receive coherent.

I. INTRODUCTION

Multichannel signals coherent processing is implemented by ADCAR, which consists of distributed platforms in the air [1]. In general, there are two broad categories of distributed radar systems for signal processing strategy, namely, distributed MIMO radar [2], [3], where the nodes are widely separated from each other, exploiting the spatial and geometric diversity of the target to enhance target detection, and DCAR [4], [5], which use the TCPs to obtain coherence gain of multichannel radar echoes thus achieving a significant improvement in target detection. Necessary coordination for TCPs must be carried out to enable the whole system to work in the TRC stage [6] due to issues, such as geometric topology and system synchronization accuracy [7], [8], [9], especially for ADCAR [10]. To address this issue, lots of solutions have been proposed, such as open-loop structures [11], [12], where the radars self-align with supplementary devices and reference points; closed-loop structures [13], [14], [15], where the target echoes are used to align the TCPs. However, the open-loop system is more demanding in terms of synchronization.

Considering the current system, this article focuses on closed-loop structures. For closed-loop structures, There are two operational stages: 1) the RC stage [16], where each unit radar emits the orthogonal signals to sense the unknown target to obtain the TCPs and complete the RC detection and 2) the TRC stage [1], [17], where the transmitting and receiving signals are adjusted using the

TCPs (velocity, delay, phase, etc., when these parameters are stable) estimated by the RC stage to achieve the performance of the large-aperture radar equivalently. Unfortunately, there is no a priori information on TCPs in most scenarios. For the TCPs, the closed-form CRB of TCPs is derived in [18] and [19], and simulation results analyze the relationship between TCPs error and SNR gain loss in the TRC stage. Liu et al. [20] proposed a clean signal echo reconstruction method to estimate unbiased TCPs under nonideal orthogonal signals. Wang et al. [21] proposed an STAP method based on stepped-frequency signals to address the complex clutter scenarios faced in the TRC stage of ADCAR and gave feasible bounds of the algorithm under nonideal factors. Since the ultimate goal of DCAR is to obtain M^3 SNR gain (M representing the number of radars) in the TRC stage, better implementing the RC stage to obtain M^2 SNR gain and higher accuracy of TCPs estimation is essential for ADCAR [22].

Recently, many scholars have also researched signal design [23], [24] and target detection [25], [26], [27] among multichannel echoes. Li et al. [28] and [29] proposed RC detector and HCD under nonideal orthogonal waveforms and analyzed the effect of TCPs error on the performance of detectors. Liu et al. [30] proposed an RC detector based on multiple clutter subspaces in ADCAR, which obtains a better moving-target detection performance than the traditional detectors. Regarding signal synthesis, analyzing the effect of phase and synchronization errors between nodes on multichannel MIMO coherent synthesis has shown that the phase information effectively improves the detection performance of ADCAR [31], [32]. However, the above works do not give a specific synthesis strategy. Although scholars have also proposed multichannel integration methods, such as the cuckoo search [33] and entropy-based methods [34] to achieve decent performance, the utilization of TCPs still needs to be improved. Further, to address the problem of coherent signal accumulation for high-speed maneuvering target detection, multichannel coherent synthesis was achieved by constructing geometrical configuration equations to solve the TCPs (calibrate distance–velocity envelope shifts and phase deviations) [35], [36], but the phase desynchronization in ADCAR is not considered. It is worth noting that distributed MIMO is mainly concerned with target detection and localization accuracy, i.e., target position and velocity, etc. In contrast, ADCAR is not only concerned with the target detection problem in the RC stage but also needs to complete the estimation of the TCPs.

ADCAR offers more significant advantages than DCAR, such as higher mobility, smarter deployment topology, etc. However, the geometrical configuration between ADCAR and the region of interest, as well as the phase desynchronization arising from physically isolated unit radar, poses severe challenges for signal coherent synthesis and TCPs estimation, including.

1) The distance–velocity envelope shifts with different properties in the complex detection environment formed by the geometrical configuration between ADCAR and the region of interest. Relative phase deviations between physically isolated unit radar and the target tend to occur, resulting in mismatches of TCPs and severely reducing the efficiency of the RC synthesis.
2) Traditional TCPs estimation methods (e.g., PP method [1], [36], [37], and CC method [1], [35], [37]) cannot obtain TCP estimation in low SNR scenario, even if the number of platforms increases. Moreover, the high-dimensional TCPs also bring enormous estimation difficulty and computational burden as the number of platforms increases.

In this article, we considered the signal coherent synthesis of airborne moving target signals problem based on multiple TR pairs for ADCAR in the RC stage. The research objectives of this article are as follows: On the one hand, the meaning of signal coherent synthesis in the RC stage is to enhance the detection performance of airborne moving targets for ADCAR. On the other hand, we propose a more efficient and accurate estimation method for TCPs than traditional algorithms. The main contributions are summarized as follows.

1) The signal model and TCPs description for ADCAR in the RC stage are established, incorporating the geometrical configuration between ADCAR and the moving target of interest in 3-D space. On this basis, a parameter space division using the DR condition for TCPs is proposed. Then, specific arithmetic examples for the DR condition in the ideal case are given to divide the probe space of interest further.
2) Multiple TR pairs joint TCPs estimation method based on MLE is proposed for ADCAR in the RC stage. We first introduce a nonclosed MLE generic solution for TCPs based on multiple TR pairs of radars. Then, the RC synthesis strategies are performed with the DR and NDR conditions. Further, two fast implementations based on PSO and CG are proposed for the RC synthesis of moving targets and TCPs output. Finally, the computational complexity is given.
3) Numerical simulations and experiments with real-measured data are provided to examine the performance of the proposed methods. First, the effect of TCPs mismatches in the RC stage are provided, and then the proposed methods are experimentally compared with some conventional existing methods. Second, we conducted detection experiments to compare the proposed and existing methods. Experiment results indicate that the performance of the proposed method for signal coherent synthesis, TCPs estimation, and target detection is significantly improved compared with the traditional methods, and the performance of the proposed methods becomes better with the increase of radar node numbers. In addition, the deployment of the proposed methods by the "cell" strategy demonstrates the effectiveness of the proposed method in multiple target scenarios.

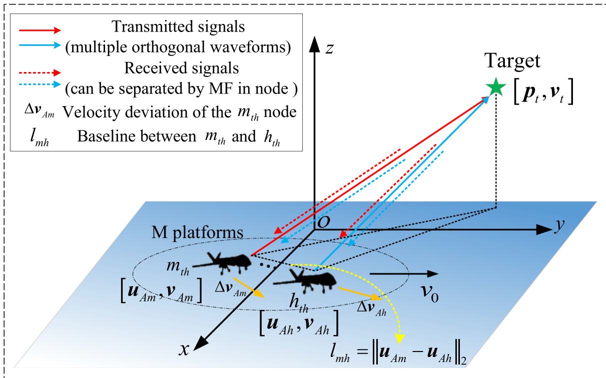

Fig. 1. ADCAR geometrical configuration in the RC stage.

The rest of this article is organized as follows: Section II presents the signal model and description of TCPs. The proposed TCPs estimation and RC methods are presented in Section III. Section IV contains two fast implementations of RC methods. Simulation and real-measured data analysis are provided in Section V. Finally, Section VI concludes this article.

Notation: In this article, the transpose is denoted by $(\cdot)^T$. Symbol $\|\cdot\|$ is the Euclidean norm. $|\cdot|$ denotes modulus of the complex number. $[a, b]$ denotes the interval from a to b. The mhth element of the vector $\mathbf{a}$ stands for $[\mathbf{a}]_{mh}$. $\mathbb{R}$ and $\mathbb{C}$ are the sets of real and complex number, respectively. round$(\cdot)$ denotes the rounding to the nearest integer. $\odot$ and $\otimes$ are defined as the Hadamard product and the Kronecker product, respectively.

II. Signal Model and TCPs Description

In this section, we first introduce the signal model and the geometry model between the ADCAR and moving target in the RC stage and then present the description of TCPs.

A. Signal Model

We consider an ADCAR system consisting of radar nodes in a 3-D Cartesian coordinate space. Without loss of generality, the radar node in the air is both a transmitter and a receiver. As shown in Fig. 1, Let $\mathbf{u}_{Am} = [x_{Am}, y_{Am}, z_{Am}]$, $\mathbf{v}_{Am} = [v^x_{Am}, v^y_{Am}, v^z_{Am}]$, $m = 1, \ldots, M$ be the position and velocity of the mth radar node. We are assuming that $\mathbf{v}_0$ is the reference velocity set for the ADCAR system, and the velocity deviation $\Delta\mathbf{v}_{Am}$ (with respect to $\mathbf{v}_0$) present in the mth platform. $\mathbf{p}_t = [x_t, y_t, z_t]$, $\mathbf{v}_t = [v_x, v_y, v_z]$ are the position and velocity of the target. The information about radar nodes is known, on the contrary, the target information is unknown.

According to the ADCAR geometrical configuration, the relative delay and velocity of the target at the mhth TR pair can be expressed as

$$\tau_{mh} = (\|\mathbf{p}_t - \mathbf{u}_{Am}\| + \|\mathbf{p}_t - \mathbf{u}_{Ah}\|)/c \tag{1}$$

$$v_{mh} = (\mathbf{v}_t - \mathbf{v}_{Am}) \cdot \frac{(\mathbf{p}_t - \mathbf{u}_{Am})}{\|\mathbf{p}_t - \mathbf{u}_{Am}\|} + (\mathbf{v}_t - \mathbf{v}_{Ah}) \cdot \frac{(\mathbf{p}_t - \mathbf{u}_{Ah})}{\|\mathbf{p}_t - \mathbf{u}_{Ah}\|} \tag{2}$$

where the velocity can be given as the normalized Doppler frequency $f_{d,mh} = v_{mh}T_r/\lambda$, and T_r is the pulse repetition interval. In the RC stage, M airborne radars transmit M orthogonal waveforms to explore areas of common interest [38]. Each radar transmits a series of K periodic pulses, over a CPI. Specifically, the target echo at the hth radar is described as

$$s_h(t) = \sum_{m=1}^{M} \alpha A_m \xi_{mh} u_m\left(t - \tau_{mh}\right) \times e^{-j2\pi f_c \tau_{mh}} e^{j2\pi f_{d,mh} t} e^{j\tilde{\psi}_{mh}} \tag{3}$$

where α is target amplitude, $u_m(t) = \sum_{k=0}^{K-1} p_m(t - kT_r)$ is the transmitted waveform of the mth radar node, $p_m(t)$ is the complex envelope of orthogonal waveform, which has the unit energy $\int_{T_p} |p_m(t)|^2 dt = 1$ and the same duration T_p, f_c is the carrier frequency, $\tilde{\psi}_{mh}$ denotes the initial phase offset, A_m is the transmit signal amplitude, and ξ_{mh} is the channel coefficient associated with the TR pair. For simplicity, it is assumed that the airborne radar operates in an upward-looking clutter-free or that the echoes have been processed based on STAP clutter suppression. Therefore, the effects of clutter are ignored. In the RC stage, each receiver has a set of M MFs, each matched to one of M orthogonal waveforms. For the hth receiver, $g_m(t) = p^*_m(-t)e^{j2\pi f_{d,mh} t}$ (MF) is used to separate the corresponding orthogonal waveform and pulse compression. Then, the TR pair can be written as

$$\begin{aligned} x_{mh}(t) &= \sum_{\bar{m}=1}^{M} \alpha A_{\bar{m}} \xi_{\bar{m}h} e^{-j2\pi f_c \tau_{mh}} e^{j\tilde{\psi}_{mh}} e^{j2\pi f_{d,mh}(t-\tau_{\bar{m}h})} \\ &\quad \times \sum_{k=0}^{K-1} \chi_{m\bar{m}}\left(t - \tau_{\bar{m}h} - kT_s, f_{\bar{m}h} - f_{mh}\right) \\ &\quad \times e^{j2\pi kT_s(f_{\bar{m}h} - f_{mh})} \\ m &= 1, \ldots, M;\ h = 1, \ldots, M;\ k = 0, \ldots K-1. \end{aligned} \tag{4}$$

The 11th TR pair is selected as the reference TR pair, and the delay range of the "cell" for the region of interest is $[\tau_c^{\min}, \tau_c^{\max}]$. The sampling interval is given as Δ_τ based on the sampling rate, and the corresponding number of echo sampling points is

$$N_\tau = \mathrm{round}\left(\frac{\tau_c^{\max} - \tau_c^{\min}}{\Delta_\tau}\right) \tag{5}$$

then, the discrete indexes of the delay can be expressed as

$$\tau_c(l) = \tau_c^{\min} + l\Delta_\tau,\ l = 1, 2, \ldots, N_\tau. \tag{6}$$

Let $t = \tau_c(l) + kT_r$ and stack (5) yields the corresponding sampling model as follows:

$$\mathbf{x}_{mh} = \alpha \mathbf{S}_{d,h} \mathbf{h}_{mh} \tag{7}$$

where

1) $\mathbf{S}_{d,h} = [\mathbf{s}(f_{d,1h}), \ldots, \mathbf{s}(f_{d,Mh})]$ is Doppler steering matrix, with $\mathbf{s}(f_d) = [1, e^{j2\pi f_d}, \ldots, e^{j2\pi(K-1)f_d}]^T$;
2) $[\mathbf{h}_{mh}]_{\bar{m}} = A_{\bar{m}}\xi_{\bar{m}h}e^{j\psi_{\bar{m}h}}\chi_{m\bar{m}}(\tau_c(l) - \tau_{\bar{m}h}, 0)$ denote the target vector at the mhth TR pair. the Doppler shift of the target is ignored, i.e., $\chi_{m\bar{m}}(\tau_c(l) - \tau_{\bar{m}h}, f_{\bar{m}h} - f_{mh}) \approx \chi_{m\bar{m}}(\tau_c(l) - \tau_{\bar{m}h}, 0)$;
3) $\chi_{m\bar{m}}(v, f) = \int p_m(u)p^*_{\bar{m}}(u - v)e^{j2\pi fu}du$ is defined as the CAF;
4) ψ_{mh} is defined as the total relative phase between the mhth TR pair and 11th pair, based on the coupling characteristics of the phase.

It is mentioned that the echo mathematical expressions and the corresponding sampling model are derived. The description of TCPs will be discussed in the following subsection.

B. TCPs Description

Based on existing methods [39] and the synchronization performance of the actual ADCAR system, it is reasonable to assume that time-frequency synchronization between each node can be guaranteed for both the transmitter and the receiver. The TCPs based on Fig. 1 are described as

$$\Delta\tau^t_m = \frac{1}{M}\sum_{h=1}^{M}(\tau_{mh} - \tau_{1h}), m = 2, \ldots, M \tag{8a}$$

$$\Delta\tau^r_h = \frac{1}{M}\sum_{m=1}^{M}(\tau_{mh} - \tau_{m1}), h = 2, \ldots, M \tag{8b}$$

$$\Delta f^t_{d,m} = \frac{1}{M}\sum_{h=1}^{M}\left(f_{d,mh} - f_{d,1h}\right), m = 2, \ldots, M \tag{8c}$$

$$\Delta f^t_{d,h} = \frac{1}{M}\sum_{m=1}^{M}\left(f_{d,mh} - f_{d,m1}\right), h = 2, \ldots, M \tag{8d}$$

$$\Delta\psi^t_m = \frac{1}{M}\sum_{h=1}^{M}(\psi_{mh} - \psi_{1h}), m = 2, \ldots, M \tag{8e}$$

$$\Delta\psi^r_h = \frac{1}{M}\sum_{m=1}^{M}(\psi_{mh} - \psi_{m1}), h = 2, \ldots, M \tag{8f}$$

where $\Delta\tau^t_m$, $\Delta f^t_{d,m}$, and $\Delta\psi^t_m$ denote the delay offset, normalized Doppler frequency offset, and phase offset generated by the mth transmitting platform, respectively, and $\Delta\tau^r_h$, $\Delta f^r_{d,h}$, and $\Delta\psi^r_h$ are defined similarly.

Obviously, the delay, phase, and Doppler can be expressed, respectively, as $\tau_{mh} = \tau_{11} + \Delta\tau^t_m + \Delta\tau^r_h$, $\psi_{mh} = \psi_{11} + \Delta\psi^t_m + \Delta\psi^r_h$, and $f_{d,mh} = f_{d,11} + \Delta f^t_{d,m} + \Delta f^r_{d,h}$. Further, the following relationship based on the synchronization assumption is given as $\Delta\tau^t_m = \Delta\tau^r_h = \Delta\tau_m$ and $\Delta f^t_{d,m} = \Delta f^t_{d,h} = \Delta f_{d,m}$. Then, the set of unknown TCPs in signal model is described as

$$\boldsymbol{\eta} = \{\Delta\boldsymbol{\tau}, \Delta\boldsymbol{f}_d, \Delta\boldsymbol{\psi}\} \in \mathbb{R}^{4(M-1)\times 1} \tag{9}$$

where $\Delta\boldsymbol{\psi} = \{\Delta\boldsymbol{\psi}^t, \Delta\boldsymbol{\psi}^r\}$, $\Delta\boldsymbol{f}_d = \{\Delta f_{d,2}, \ldots, \Delta f_{d,M}\}$, and $\Delta\boldsymbol{\tau} = \{\Delta\tau_2, \ldots, \Delta\tau_M\}$. In fact, the complex detection environment formed by the geometrical configuration between ADCAR and the region of interest causes different dimensional features of the TCPs. Therefore, TCPs estimation and RC methods are discussed in the following section.

III. TCPS ESTIMATION AND RC METHODS

The geometrical configuration between ADCAR and the unknown target, i.e., the region of interest, always causes distance–velocity envelope shifts with different properties. In this section, a nonclosed MLE generic solution for TCPs based on multiple TR pairs is derived, and a DR condition for TCPs is proposed. Then, the signal coherent synthesis strategies in the RC stage are performed with the DR and NDR conditions, as shown in Fig. 2.

A. Multiple TR Pairs Joint TCPs Estimation

We consider sampling model (7) for ADCAR in the RC stage, shown in the block named separated multichannel echoes in Fig. 2. The mhth TR pair can be written as

$$\mathbf{y}_{mh} = \mathbf{x}_{mh} + \mathbf{w}_{mh} \tag{10}$$

where $\mathbf{w}_{mh}$ is denoted as the noise, which is assumed to follow a zero-mean complex Gaussian distribution and to be mutually uncorrelated, independent and identically distributed among different receivers and pulses. Then, the sampling model for all TR pairs can be rewritten as

$$\begin{aligned}\mathbf{y} &= \begin{bmatrix}\mathbf{y}_{11}\\ \vdots \\ \mathbf{y}_{m1} \\ \vdots \\ \mathbf{y}_{MM}\end{bmatrix} = \begin{bmatrix}\mathbf{S}_{11} & & & & \\ & \ddots & & & \\ & & \mathbf{S}_{m1} & & \\ & & & \ddots & \\ & & & & \mathbf{S}_{MM}\end{bmatrix}\mathbf{1}_{MM} + \mathbf{w}\\ &= \alpha\mathbf{U}\mathbf{1}_{MM} + \mathbf{w}\end{aligned} \tag{11}$$

where $\mathbf{S}_{mh} = \begin{bmatrix}\mathbf{s}_{mh}(0)\\ \vdots \\ \mathbf{s}_{mh}(K-1)\end{bmatrix}$ and $\mathbf{s}_{mh}(k) = [s^k_{m,1h}, s^k_{m,2h}, \ldots, s^k_{m,Mh}]$, and $s^k_{m,\bar{m}h}$ is defined as

$$s^k_{m,\bar{m}h} \approx A_{\bar{m}}\xi_{\bar{m}h}e^{j\psi_{\bar{m}h}}e^{j2\pi kf_{d,\bar{m}h}}\chi_{m\bar{m}}\left(\tau_c(l) - \tau_{\bar{m}h}, 0\right). \tag{12}$$

For the set of unknown TCPs $\boldsymbol{\eta}$, the log-likelihood function of the echoes is given as

$$\begin{aligned}L(\mathbf{y}\,|\boldsymbol{\eta}) = &-(\mathbf{y} - \alpha\mathbf{U}_{MM}\mathbf{1}_{MM})^H\mathbf{R}^{-1}(\mathbf{y} - \alpha\mathbf{U}\mathbf{1}_{MM})\\ &-\ln|\mathbf{R}| - C\end{aligned} \tag{13}$$

where $\mathbf{R} = \sigma^2_w\mathbf{I}_{M^2KL_s}$ is the noise covariance, and σ^2_w is the variance. Based on the derivative of (13), the MLE of α is given by

$$\hat{\alpha} = \frac{(\mathbf{U}\mathbf{1}_{MM})^H\mathbf{R}^{-1}\mathbf{y}}{(\mathbf{U}\mathbf{1}_{MM})^H\mathbf{R}^{-1}(\mathbf{U}\mathbf{1}_{MM})}. \tag{14}$$

Substituting (14) into (13) and omitting the irrelevant terms and the normalized denominator, the MLE generic solution

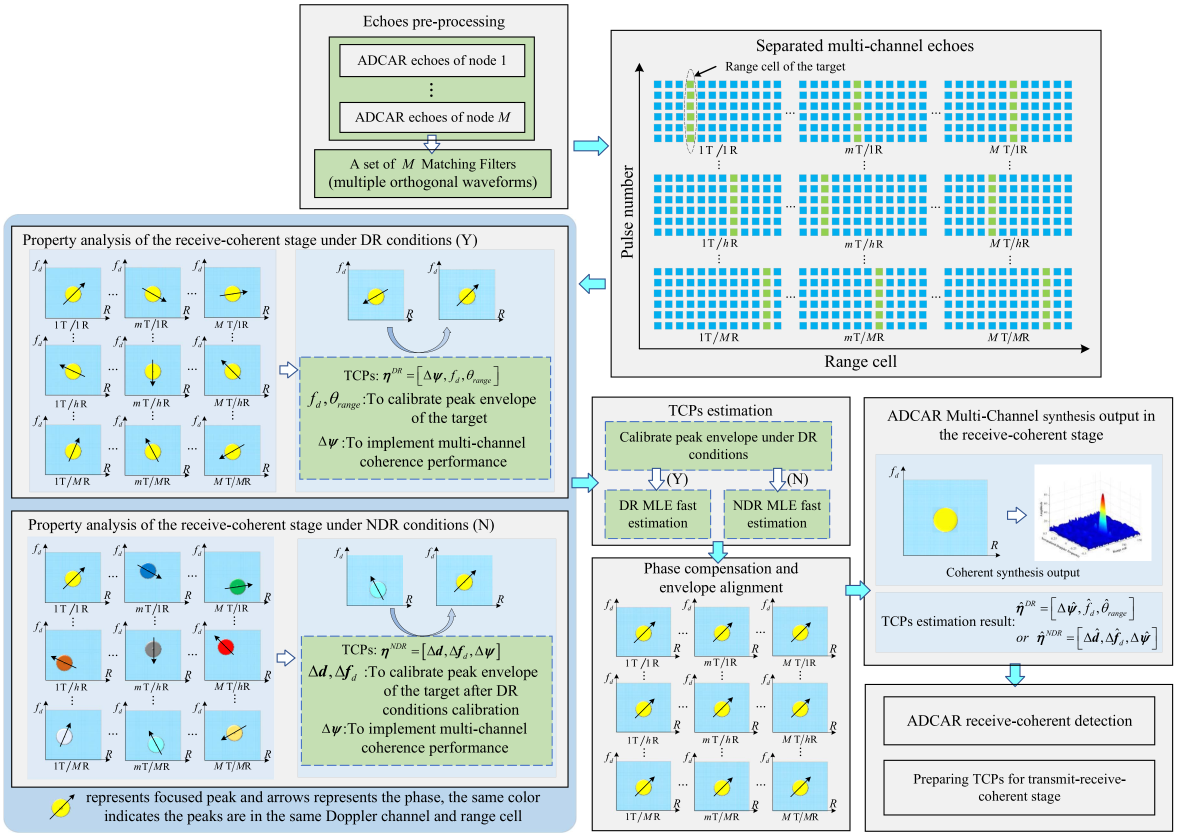

Fig. 2. Flowchart of the proposed signal coherent synthesis strategies in the RC stage

for TCPs can be determined by

$$\hat{\boldsymbol{\eta}}_{ML} = \arg\min_{\boldsymbol{\eta}} \left\{ -\left| \sum_{h=1}^{M} \sum_{k=0}^{K-1} \sum_{m=1}^{M} \left(y_{mh}^{k}\right)^{H} \sum_{\bar{m}=1}^{M} s_{m,\bar{m}h}^{k} \right|^{2} \right\}. \tag{15}$$

Concerning (15), there is no closed-form solution for the MLE estimation of TCPs, and it is difficult to avoid the joint search for multidimensional parameters. Fortunately, the dimensionality of the TCPs is reduced from $3M^2$ to $4(M-1)$ by the description of TCPs analyzed in Section II. However, the computational complexity of multidimensional parameter search is still not negligible.

Remark 1: TCPs are primarily designed to calibrate the differences between different TR pairs, while the 1-D unknown parameters of the reference TR pair, such as τ_{11}, can be obtained based on the synthesis results through conventional detection.

B. DR Condition

The target signal coherence criterion [40] divides the coherent detection region of ADCAR based on the baseline length, signal wavelength, target size, and detection distance. On this basis, we proposed a DR condition for TCPs to divide the probe space of interest. Specifically, the geometrical configuration noted as the DR condition is satisfied by utilizing low-dimensional TCPs to achieve signal coherent synthesis without significant performance loss; the opposite is noted as the NDR condition.

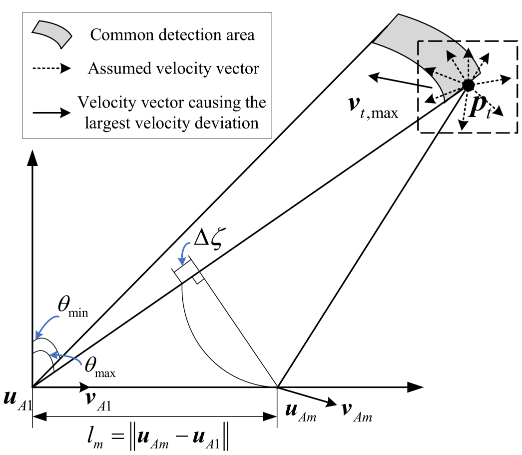

Fig. 3. Schematic of the geometric configuration between the dual nodes and the region of interest

As shown in Fig. 3, the angle range of the "cell" for the region of interest is $[\theta_{\min}, \theta_{\max}]$, and the baseline between platforms is expressed as l_m. Based on the geometrical configuration and signal coherent synthesis efficiency, the DR condition for TCPs is described as

$$\frac{2\pi}{\lambda} \left| \|\mathbf{p}_t - \mathbf{u}_{Am}\| - \|\mathbf{p}_t - \mathbf{u}_{A1}\| - l_m \sin\theta_{\max} \right| < \varepsilon_\tau \tag{16}$$

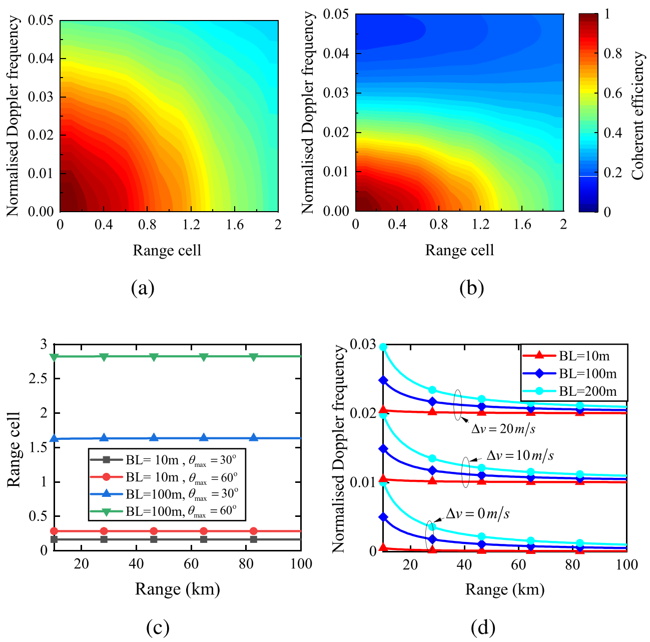

Fig. 4. Specific arithmetic simulation with the DR condition in the RC stage. (a) and (b) are the relationship between different deviations and coherent synthesis efficiency for $K = 8$ and $K = 16$, respectively. (c) and (d) show the result for the different deviations versus range in different scenarios

$$\frac{T_r}{\lambda}\left|\left(\mathbf{v}_{t,\max}-\mathbf{v}_{A1}\right)\cdot\left(\frac{(\mathbf{p}_t-\mathbf{u}_{A1})}{\|\mathbf{p}_t-\mathbf{u}_{A1}\|}-\frac{(\mathbf{p}_t-\mathbf{u}_{Am})}{\|\mathbf{p}_t-\mathbf{u}_{Am}\|}\right)\right.$$
$$\left.+\,\Delta\mathbf{v}_{Am}\cdot\frac{(\mathbf{p}_t-\mathbf{u}_{Am})}{\|\mathbf{p}_t-\mathbf{u}_{Am}\|}\right|<\varepsilon_f \quad (17)$$

where $\mathbf{v}_{t,\max}$ denotes the maximum target velocity vector generated relative to the reference node, i.e., the velocity vector generated by the target moving in any direction at an assumed maximum scalar velocity $v_{t,\max}$. $\Delta\mathbf{v}_{Am}$ is the deviation vector of the mth node for the reference node. ε_τ and ε_f are defined as the tolerance of Doppler frequency deviation and range cell deviation, respectively, and their values are related to the signal coherent synthesis efficiency. The signal coherent synthesis efficiency can be defined as

$$\kappa = \frac{\mathrm{SNR}_{\mathrm{ch}}}{KM^2\mathrm{SNR}_s} \quad (18)$$

where $\mathrm{SNR}_{\mathrm{ch}}$ and SNR_s are the average SNR of the coherent synthesis output and the single monopulse echo, respectively. Obviously, $0 \le \kappa \le 1$ and optimal coherent synthesis KM^2 gain for single monopulse echo.

We give the basis for the corresponding simulation example in Fig. 4. Fig. 4(a) and (b) show the effect of Doppler frequency and range cell deviation on the signal coherent synthesis efficiency. The interval of the corresponding deviation can be obtained by limiting the coherent synthesis efficiency, such as $\kappa \ge 0.9$. Then, we calculate the deviations in the geometrical configuration between ADCAR and the region of interest to determine whether the detection scenario satisfies the DR condition, as shown in Fig. 4(c) and (d).

Therefore, we can divide the detection region by the DR condition to further select the RC synthesis strategy, as shown in Fig. 2. The following subsections give the signal coherent synthesis strategies in the RC stage.

C. RC Synthesis With DR Condition

For simplicity, it is assumed that the transmitted waveforms are ideal orthogonal waveforms. Equation (4) can be rewritten as

$$x_{mh}(t) = \alpha A_m \xi_{mh} e^{j\psi_{mh}} \times \sum_{k=0}^{K-1} \chi\left(t-\tau_{mh}-kT_s, 0\right) e^{j2\pi kT_s f_{d,mh}}. \quad (19)$$

When the geometrical configuration satisfies the DR condition, it is clear that $\Delta\tau_m = \frac{l_m \sin\theta}{\lambda}$, where θ is the angle between the target and the direction normal to the 1st platform. Similarly, let $t = \tau_c(l) + kT_s$, the target echo vector in the DR condition could be written as

$$\mathbf{x} = \tilde{\alpha}\left(\mathbf{s}_d\left(f_d\right) \otimes \left(\mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) \otimes \mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right) \odot \boldsymbol{a}\left(\tau_c(l), \Delta\boldsymbol{\tau}\right)\right)\right). \quad (20)$$

The complex amplitude of the target is $\tilde{\alpha} = \alpha A_1 \xi_{11} e^{j\psi_{11}}$, $\boldsymbol{l}_B = [l_2, \ldots l_M]^T$ is the set of baselines, $\mathbf{s}_d(f_d) = [1, \ldots, e^{j2\pi kT_r f_d}, \ldots, e^{j2\pi(K-1)T_r f_d}]^T$ is the Doppler steering vector, the pulse compression outputs are $[\boldsymbol{a}(\tau_c(l), \Delta\boldsymbol{\tau})]_{mh} = \chi(\tau_c(l) - \tau_{11} - \Delta\tau_m - \Delta\tau_h, 0)$. The phase compensation vector between each TR pair can be expressed as

$$\mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) = \left[1, \ldots, e^{j\pi(m-1)\Delta\psi_m^t}, \ldots, e^{j\pi(M-1)\Delta\psi_M^t}\right]^T$$
$$\mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right) = \left[1, \ldots, e^{j\pi(m-1)\Delta\psi_m^r}, \ldots, e^{j\pi(M-1)\Delta\psi_M^r}\right]^T. \quad (21)$$

As shown in Fig. 2, the target's envelope shifts and relative phase deviations of different TR pairs need to be aligned to achieve the RC synthesis. Assuming that $[\theta_{\min}, \theta_{\max}] \in [-\pi/2, \pi/2]$ and the baseline set $\boldsymbol{l}_B$ is deterministic. Let $\rho \in [\sin\theta_{\min}, \sin\theta_{\max}]$ and alignment of the envelope at different angles can be accomplished by a 1-D search of ρ. The number of envelope shift points for the mhth TR pair can be written as

$$\Delta d_{mh}^{\mathrm{DR}} = \mathrm{round}\left(\frac{l_m \rho}{cT_s} + \frac{l_h \rho}{cT_s}\right) \quad (22)$$

where c is the velocity of light and T_s is the sampling interval. The step size of ρ is related to the sampling rate. It can be found that the distance domain envelope alignment is achieved by considering the target as being at the current sampling point when the target angle is indistinguishable at the sampling interval. The corresponding echo vector can be expressed as

$$\mathbf{x} = \tilde{\alpha}\left(\mathbf{s}_d\left(f_d\right) \otimes \left(\mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) \otimes \mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right) \odot \boldsymbol{a}\left(\tau_c(l), \rho\right)\right)\right) \quad (23)$$

where $[\boldsymbol{a}(\tau_c(l),\rho)]_{mh} = \chi(\tau_c(l) - \tau_{11} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau, 0)$. The TCPs with the DR condition can be expressed as

$$\boldsymbol{\eta}^{\mathrm{DR}} = \{\Delta\boldsymbol{\psi}\} \in \mathbb{R}^{2(M-1)\times 1}. \tag{24}$$

Based on (15), the TCPs estimator with the DR condition can be expressed as

$$\hat{\boldsymbol{\eta}}_{\mathrm{ML}}^{\mathrm{DR}} = \arg\min_{\boldsymbol{\eta}^{\mathrm{DR}}} \left\{ -\left| \sum_{h=1}^{M}\sum_{k=0}^{K-1}\sum_{m=1}^{M} \left(y_{mh}^{k}(\tau_c(l))\right)^H s_{mh}^{k} \times \chi\left(\tau_c(l) - \tau_{11} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau, 0\right)\right|^2 \right\} \tag{25}$$

where $s_{mh}^{k} = e^{j\Delta\psi_m^t} e^{j\Delta\psi_h^r} e^{j2\pi k T_r f_d}$ is the matching compensation term, y_{mh}^{k} denotes the sampling echo of the kth pulse of the mhth TR pair with the DR condition. With the matching compensation and the alignment of the envelope, the signal coherent synthesis strategies in the RC stage with the DR condition could be expressed as

$$\begin{aligned}\Upsilon_{\mathrm{DR}}&\left(\tau_c(l), \rho, f_d, \hat{\boldsymbol{\eta}}_{\mathrm{ML}}^{\mathrm{DR}}\right)\\ &\approx \sum_{h=1}^{M}\sum_{k=0}^{K-1}\sum_{m=1}^{M} \left(y_{mh}^{k}(\tau_c(l))\right)^H s_{mh}^{k}\\ &\quad\times \chi\left(\tau_c(l) - \tau_{11} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau, 0\right).\end{aligned} \tag{26}$$

Note that the delay envelope of the target between different TR pairs in the DR condition is completed by ρ 1-D search. Thus, the TCPs characterizing the differences between TR pairs obtain a DR for ρ, f_d in the DR condition. In addition, (25) demonstrates that the output of TCPs results from the multiple TR pairs joint.

D. RC Synthesis With NDR Condition

When the geometrical configuration between ADCAR and the region of interest does not satisfy the DR condition, i.e., With the NDR condition, the target echo vector could be written as

$$\begin{aligned}\mathbf{x} &= \tilde{\alpha}\mathbf{s}_d\left(f_d\right)\\ &\quad\otimes\left(\mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right)\otimes\mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right)\odot\boldsymbol{a}\left(\tau_c(l),\rho,\Delta\boldsymbol{d}\right)\right)\odot\Delta\mathbf{s}_d\end{aligned} \tag{27}$$

where $\Delta\boldsymbol{d}$ denotes the number of envelope shifts sampling points, and the corresponding range cell envelope can be expressed as

$$\begin{aligned}&\left[\boldsymbol{a}\left(\tau_c(l),\rho,\Delta\boldsymbol{d}\right)\right]_{mh}\\ &= \chi\left(\tau_c(l) - \tau_{11} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau - \Delta d_m - \Delta d_h, 0\right).\end{aligned} \tag{28}$$

$\Delta\mathbf{s}_d$ is defined as the deviations of Doppler steering vector between different TR pairs and can be expressed as

$$\Delta\mathbf{s}_d = \left[\mathbf{1}_{1\times K},\ldots,\mathbf{s}_d\left(\Delta f_{d,m1}\right)^T,\ldots,\mathbf{s}_d\left(\Delta f_{d,MM}\right)^T\right]^T. \tag{29}$$

As shown in Fig. 2, the implementation of RC synthesis with the NDR condition requires adding deviation parameters to RC strategy with the DR condition. The unknown TCPs of ADCAR with NDR conditions can be presented as

$$\boldsymbol{\eta}^{\mathrm{NDR}} = \left[\Delta\boldsymbol{d}, \Delta\boldsymbol{f}_d, \Delta\boldsymbol{\psi}\right] \in \mathbb{R}^{4(M-1)\times 1} \tag{30}$$

and the number of envelope shifts sampling points satisfy $|\Delta d_{mh}| \le$ found $(|\frac{|l_m - l_h|}{cT_s} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau|)$. Compared to (15), there is a reduction in the number of search points, further relieving the computational burden of multidimensional parameter searching. Thus, the TCPs estimator with the NDR condition can be expressed as

$$\hat{\boldsymbol{\eta}}_{\mathrm{ML}}^{\mathrm{NDR}} = \arg\min_{\boldsymbol{\eta}^{\mathrm{NDR}}} \left\{ -\left| \sum_{h=1}^{M}\sum_{k=0}^{K-1}\sum_{m=1}^{M} \left(\tilde{y}_{mh}^{k}(\tau_c(l))\right)\tilde{s}_{mh}^{k} \times \chi\left(\tau_c(l) - \tau_{11} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau - \left(\Delta d_m + \Delta d_h\right)\Delta_\tau, 0\right)\right|^2 \right\} \tag{31}$$

where $\tilde{s}_{mh}^{k} = e^{j\Delta\psi_m^t} e^{j\Delta\psi_h^r} e^{j2\pi k T_r f_d} e^{j2\pi k T_r \Delta f_{mh}}$ is the matching compensation term in the NDR condition, $\tilde{y}_{mh}^{k}$ denotes the sampling echo of the kth pulse of the mhth TR pair with the NDR condition. After the matching compensation and the alignment of the envelope, the signal coherent synthesis strategies in the RC stage with the NDR condition could be expressed as

$$\begin{aligned}\Upsilon_{\mathrm{NDR}}&\left(\tau_c(l), \rho, f_d, \hat{\boldsymbol{\eta}}_{ML}^{\mathrm{NDR}}\right)\\ \approx&\sum_{h=1}^{M}\sum_{k=0}^{K-1}\sum_{m=1}^{M} \left(\tilde{y}_{mh}^{k}(\tau_c(l))\right)^H \tilde{s}_{mh}^{k}\\ &\times \chi\left(\tau_c(l) - \tau_{11} - \Delta d_{mh}^{\mathrm{DR}}\Delta_\tau - \left(\Delta d_m + \Delta d_h\right)\Delta_\tau, 0\right).\end{aligned} \tag{32}$$

It is worth noting that we divide the region of interest into multiple "cell" in actual target detection. This "cell" is generally generated according to different parameter spaces, such as the "cell" in this article consists of $[\tau_c^{\min}, \tau_c^{\max}]$, $[f_d^{\min}, f_d^{\max}]$, and $[\theta_{\min}, \theta_{\max}]$, and assumes that there is at most one target in that "cell." The reason for this assumption is that the baseline of the ADCAR tends not to be too long so that the geometrical configuration between ADCAR and the region of interest does not cause a huge distance–velocity envelope shift. Therefore, the strategy in this article has still been able to achieve RC synthesis in multitarget scenarios.

IV. FAST IMPLEMENTATIONS OF RC METHODS

As the number of nodes increases, the grid search of TCPs imposes an exponential computational burden and grid mismatch error for ADCAR. To further accelerate the implementations of the proposed RC strategies, two fast implementations of RC methods based on PSO and CG are demonstrated for the signal coherent synthesis of airborne moving targets and TCPs output in this section.

A. PSO Mehthod

The PSO [41], [42] algorithm can be described as extreme value optimization by particle fitness, which is easy to implement and requires low computational resources. The unknown parameter set is invested with multiple particles, i.e., multiple stochastic solutions are initialized, and then the global optimal solution is found by the fitness of the particle swarm. However, the particle swarm may fall into the local optima. Fortunately, the TCPs-MLE estimator (15) is only likely to have at most one optimal value under the “cell” assumption. Since the a priori information about the target’s velocity is unknown, the FFT can quickly search the Doppler frequency domain. Therefore, the following steps propose a TCPs-MLE estimation method (PSO-FFT-MLE) with the DR and NDR conditions.

Step 1 (Initialization): Initialize PSO iteration parameters (the particle number I, the maximum iteration number $j_{\max}$, the tolerance γ). Initialize the TCPs vectors (position) $\boldsymbol{\eta}_{(0),i}(\boldsymbol{\eta}^{\mathrm{DR}}_{(0),i}$ or $\boldsymbol{\eta}^{\mathrm{NDR}}_{(0),i}$ and the velocity vectors $\boldsymbol{v}_i^{(0)}$ of each particle, where $i = 1, 2, \ldots, I$ is the particle index number. The particle number I should satisfy $I \ll L_{\mathrm{grid}} = \prod_{m=1}^{M^2} L_m$, where L_m is the number of grids in the mth dimension and L_{grid} denotes the number of grids in the TCPs for ergodic search.

Step 2 (Calculating Fitness Function): For the jth iteration, calculate the fitness functions of I. For (25) and (32), the corresponding fitness function is given by

$$\begin{aligned} & f^{\mathrm{DR}}_{fit}\left(\boldsymbol{\eta}^{\mathrm{DR}}_{(j),i}\right) \\ & = \min\left\{-\left|\mathbf{s}_d\left(f_d\right)^H\left(\mathbf{I}_K \otimes \mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) \otimes \mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right)\right)^H\mathbf{y}\right|^2\right\} \\ & \qquad \text{s.t.} \quad f_{d,\min} \le f_d \le f_{d,\max} \end{aligned} \tag{33}$$

$$\begin{aligned} & f^{\mathrm{NDR}}_{fit}\left(\boldsymbol{\eta}^{\mathrm{NDR}}_{(j),i}\right) \\ & = \min\left\{-\left|\mathbf{s}_d\left(f_d\right)^H\left(\mathbf{I}_K \otimes \mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) \otimes \mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right)\right)^H\right.\right. \\ & \left.\left.\left(\Delta\mathbf{s}_d^H \odot \tilde{\mathbf{y}}\right)\right|^2\right\} \\ & \text{s.t.} \quad f_{d,\min} \le f_d \le f_{d,\max} \end{aligned} \tag{34}$$

where $j = 1, \ldots, j_{\max}$ is iteration number and $\Delta\psi^j_{i,1} = 1$, the phase compensation vector is $\mathbf{a}(\Delta\boldsymbol{\psi}) = [1, \ldots e^{j\Delta\psi^j_{i,M}}]^T$. $\mathbf{y}$, $\tilde{\mathbf{y}}$ denote the echo vectors at the corresponding sample time with the DR and NDR conditions, respectively, and their components can be expressed as

$$[\mathbf{y}]_{mhk} = y^k_{mh}(\tau_c(l))\chi\left(\tau_c(l) - \tau_{11} - \Delta d^{\mathrm{DR}}_{mh}\Delta_\tau, 0\right) \tag{35}$$

$$\begin{aligned} [\tilde{\mathbf{y}}]_{mhk} = & \tilde{y}^k_{mh}(\tau_c(l))\chi\left(\tau_c(l) - \tau_{11} - \Delta d^{\mathrm{DR}}_{mh}\Delta_\tau\right. \\ & \left. - \left(\Delta d_m + \Delta d_h\right)\Delta_\tau, 0\right). \end{aligned} \tag{36}$$

Step 3 (Updating Global and Personal Best Position): Update the global best position $\boldsymbol{\eta}_{(j),g}$ and the personal best positions $\boldsymbol{\eta}_{(j),p,i}$

$$\boldsymbol{\eta}_{(j),g} = \begin{cases} \boldsymbol{\eta}_{(j),p,i} f_{fit}\left(\boldsymbol{\eta}_{(j),p,i}\right), & \ge f_{fit}\left(\boldsymbol{\eta}_{(j-1),p,i}\right) \\ \boldsymbol{\eta}_{(j-1),g}, & \text{else} \end{cases}$$

$$\boldsymbol{\eta}_{(j),p,i} = \begin{cases} \boldsymbol{\eta}_{(j),i}, & f_{fit}\left(\boldsymbol{\eta}_{(j),i}\right) \ge f_{fit}\left(\boldsymbol{\eta}_{(j-1),p,i}\right) \\ \boldsymbol{\eta}_{(j-1),p,i}, & \text{else.} \end{cases} \tag{37}$$

Algorithm 1: PSO-FFT-MLE Algorithm.

```
Obtain the echo of TR pair y^k_mh(τ_c(l)) in (4).
Divide the region of interest into "cell" with DR
   condition or not via (16) and (17).
Initialize TCPs vectors, i.e., η_(0),i(η^DR_(0),i or η^NDR_(0),i)
   and v_i^(0). Initialize PSO parameters, i.e., I, j_max, γ.
   For j = 1, ..., j_max
      Obtain y or ỹ via (35), (36).
      Calculate f_fit(η_(j),i) via (33), (34).
      Update η_(j),g and η_(j),p,i via (37).
      Update v_i^(j+1) and η_(j+1),i via (38).
      If |f_fit(η_(j),g) − f_fit(η_(j−1),g)| < γ, break.
   End
Return η_(j),g as η̂
```

Step 4 (Updating the Velocity and Position Vectors): Update the velocity vector $\mathbf{v}_i^{(j)}$ and position vector $\boldsymbol{\eta}_{(j),i}$,

$$\begin{aligned} \mathbf{v}_i^{(j+1)} = & w_j\mathbf{v}_i^{(j)} + c_1 r_1\left(\boldsymbol{\eta}_{(j),p,i} - \boldsymbol{\eta}_{(j),i}\right) \\ & + c_2 r_2\left(\boldsymbol{\eta}_{(j),g} - \boldsymbol{\eta}_{(j),i}\right) \\ \boldsymbol{\eta}_{(j+1),i} = & \boldsymbol{\eta}_{(j),i} + \mathbf{v}_i^{(j+1)} \end{aligned} \tag{38}$$

where r_1 and r_2 are two independent variables that are randomly distributed within the range of (0,1], c_1 and c_2 are the learning coefficients, w_j is the inertia coefficient of the jth iteration, which is closely related to the convergence rate and generally decreases linearly during the iteration.

Step 5 (Residual Update): Repeat Steps 2–4 until maximum iteration number $j_{\max}$ or $|f_{fit}(\boldsymbol{\eta}_{(j),g}) - f_{\mathrm{fit}}(\boldsymbol{\eta}_{(j-1),g})| < \gamma$. Then, return $\boldsymbol{\eta}_{(j),g}$ as $\hat{\boldsymbol{\eta}}$.

By the implementation of the above algorithm, the corresponding TCPs estimates $\boldsymbol{\eta}^{\mathrm{DR}}_{(j),g}$ or $\boldsymbol{\eta}^{\mathrm{NDR}}_{(j),g}$ can be can be quickly obtained and used to compensate for the RC synthetics. Finally, we summarize the proposed PSO-FFT-MLE in Algorithm 1 for clarity.

B. CG Mehthod

The PSO-type optimization is approximately black-box, lacks a complete theoretical analysis, and is computationally burdensome to solve iteratively with multiple particles. Therefore, we further develop faster and theoretically interpretable method for DR condition. We consider the following optimization problem:

$$\min_{\boldsymbol{\eta}^{\mathrm{DR}}} f\left(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y}\right) \tag{39}$$

where $f(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y})$ denotes the optimization function of the unknown TCPs with respect to the echoes, which can be expressed as

$$f\left(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y}\right) = -\left|\mathbf{s}_d\left(f_d\right)^H\left(\mathbf{I}_K \otimes \mathbf{s}_{tr}\left(\Delta\boldsymbol{\psi}^t, \Delta\boldsymbol{\psi}^r\right)\right)^H\mathbf{y}\right|^2. \tag{40}$$

Optimization problem (39) is a nonlinear programming problem, and we use a CG method to find a local maximum to the problem. A CG method searches in directions conjugate to the cost functions gradient [43]. Thus, an iterative

Algorithm 2: CG-FFT-MLE Algorithm.

1: Obtain the echo of TR pair $y_{mh}^{k}(\tau_c(l))$ in (4).
2: Determine the "cell" satisfies the DR conditions via (16) and (17).
3: Obtain $\mathbf{y}$ via (35). Initialize parameters, i.e., $\hat{\boldsymbol{\eta}}_0^{\mathrm{DR}}$, β_0, the tolerance ε.
4: **For** $n = 1, \ldots, n_{\max}$
5: Calculate $\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}}$ via (41a).
6: Update β_n and $\boldsymbol{s}_n$ via (41b) and (41c).
7: Search α_n via (41d).
8: Update $\nabla\hat{\boldsymbol{\eta}}_{n+1}^{\mathrm{DR}}$ via (41e).
9: **If** $|f(\boldsymbol{\eta}_{n+1}^{\mathrm{DR}}|\mathbf{y}) - f(\boldsymbol{\eta}_n^{\mathrm{DR}}|\mathbf{y})| < \varepsilon$, **break**.
10: **End**
11: Return $\hat{\boldsymbol{\eta}}_{n+1}^{\mathrm{DR}}$ as $\boldsymbol{\eta}^{\hat{\mathrm{DR}}}$

scheme is depicted as follows:

$$\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}} = \nabla_{\boldsymbol{\eta}^{\mathrm{DR}}} f\left(\boldsymbol{\eta}_n^{\mathrm{DR}}|\mathbf{y}\right) \tag{41a}$$

$$\beta_n = \frac{\left(\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}}\right)^T\left(\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}} - \nabla\hat{\boldsymbol{\eta}}_{n-1}^{\mathrm{DR}}\right)}{\left(\nabla\hat{\boldsymbol{\eta}}_{n-1}^{\mathrm{DR}}\right)^T\left(\nabla\hat{\boldsymbol{\eta}}_{n-1}^{\mathrm{DR}}\right)} \tag{41b}$$

$$\boldsymbol{s}_n = -\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}} + \beta_n\boldsymbol{s}_{n-1} \tag{41c}$$

$$\alpha_n = \min_{\alpha} f\left(\boldsymbol{\eta}_n^{\mathrm{DR}} + \alpha\boldsymbol{s}_n|\mathbf{y}\right) \tag{41d}$$

$$\nabla\hat{\boldsymbol{\eta}}_{n+1}^{\mathrm{DR}} = \nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}} + \alpha_n\boldsymbol{s}_n \tag{41e}$$

where $\nabla_{\boldsymbol{\eta}^{\mathrm{DR}}} f(\boldsymbol{\eta}_n^{\mathrm{DR}}|\mathbf{y})$ is the gradient of $f(\boldsymbol{\eta}_n^{\mathrm{DR}}|\mathbf{y})$ with respect to the TCPs for iteration n, denoted by $\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}}$. The equation of β_n is chosen according to the Polak–Ribière formula [44]. The algorithm is initialized with $\hat{\boldsymbol{\eta}}_0^{\mathrm{DR}}$. A derivation of $\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}}$ is presented in the Appendix A. we summarize the proposed CG-FFT-MLE in Algorithm 2 for clarity.

Remark 2: Both proposed fast implementations are essentially multi-TR pairs joint RC synthesis methods. The PSO-FFT-MLE benefited from the advantages of intelligent optimization algorithms for combinatorial optimization problems with multidimensional parameters that can be used in DR and NDR conditions. The CG-FFT-MLE developed for DR conditions obtains faster convergence results with theoretical interpretability. As can be seen in Section V, both of them have advantages and disadvantages in terms of performance (e.g., TCPs estimation, target detection, running time). In engineering applications, ADCAR under different scenarios (e.g., computational capacity, number of platforms, and performance requirement) can be deployed with different methods, even both of them are used in conjunction to maximize performance.

C. Computational Complexity Analysis

In this section, the computational burden of the proposed methods is provided. The complexity of one complex addition is $c_a = I_{\mathrm{ca}}$, and the complexity of one complex multiplication complexity is $c_m = I_{\mathrm{cm}}$, i.e., $I_{\mathrm{cm}} = 3I_{\mathrm{ca}}$. The Doppler domain sampling points can be realized by the N_{FFT} points. In the DR condition, the number of search grid points for each phase parameter is L_φ. In the NDR condition, the number of search grid points for $\{\Delta\boldsymbol{d}, \Delta\boldsymbol{f}_d, \Delta\boldsymbol{\psi}\}$ are L_d, $L_{\Delta f}$, and L_φ, respectively. For the PSO-MLE-FFT, it is assumed that the maximum number of iterations is $j_{\max,\mathrm{DR}}$, $j_{\max,\mathrm{NDR}}$, respectively. The maximum number of iterations for CG-MLE-FFT is j_{CG}. Table I presents the computational complexities of the methods.

According to Table I, the estimation of TCPs in the DR condition has less computational complexity compared to the NDR condition. That is benefited from the DR of TCPs by the DR condition. The computational complexity of the parametric grid-based MLE-S (MLE-Search) estimator increases exponentially with the number of nodes. In contrast, the computational complexity of the two proposed off-grid fast implementation methods with guaranteed performance is lower than MLE-S, related to the corresponding iterative methods.

Noted that the PSO method is performed with multiple particles iterating simultaneously. In contrast, the CG method does not have multiple particles iterating simultaneously and only optimizes along the direction of decreasing CG of the initial value. Therefore, the computational complexity of the CG method is smaller than that of the PSO method in practice. Further, the numerical simulation results for the corresponding running times are given in Section V-D. The two proposed off-grid fast implementation methods can lessen the computational load.

V. SIMULATION AND REAL-MEASURED DATA ANALYSIS

This section aims to evaluate the performance of the proposed methods via simulation experiments and real-measured data. For comparison, some traditional methods are performed, for example, the PP-based and CC-based methods, and the traditional methods are improved by FFT to accumulate the echoes within a single channel, denoted as PP-FFT and CC-FFT, respectively.

Before that, some definitions and simulation parameter choices are given. The SNR is defined as

$$\mathrm{SNR} = \frac{|\alpha|^2}{\sigma_n^2} \tag{42}$$

where the noise variance is chosen as $\sigma_n^2 = 1$, and α is the target complex amplitude. The RC synthesis essentially expects radar echo to be coherently synthesized at the receiver. However, because of the 2π-phase period, even if there is a large difference between the estimated and actual phase values, its use for RC synthesis could present better performance. Therefore, a new definition is given for measuring the estimation performance of the phase parameter, i.e., the phase distance, whose corresponding equation can be described as

$$P_{\mathrm{distance}} = \left\| e^{j\hat{\psi}} - e^{j\psi_{\mathrm{real}}} \right\| \tag{43}$$

where $\hat{\psi}$ is the estimated phase value, ψ_{real} is the actual phase value. The root means square error of the phase

TABLE I
Computational Complexity of Different Methods

Condition	Methods	Computational Complexity
DR	MLE-S	$L_\varphi^{2M-2}\left(\left(I_{ca}+I_{cm}\right)M^2-I_{ca}\right)+M^2N_{FFT}\left(\left(I_{ca}+I_{cm}\right)K-I_{ca}\right)$
	PSO-FFT-MLE	$j_{\max,DR}I_{DR}\left(\left(I_{ca}+I_{cm}\right)M^2-I_{ca}\right)+\left(2I_{cm}-I_{ca}\right)M^2N_{FFT}\log_2\left(N_{FFT}\right)$
	PCG-FFT-MLE	$j_{CG}\left(L_a\left(\left(I_{ca}+I_{cm}\right)M^2-I_{ca}\right)+2\left(I_{ca}+I_{cm}\right)M^2 +2\left(4I_{ca}+3I_{cm}-2\right)M-2I_{ca}\right)+\left(2I_{cm}-I_{ca}\right)M^2N_{FFT}\log_2\left(N_{FFT}\right)$
NDR	MLE-S	$L_\varphi^{2(M-1)}L_d^{M-1}L_{\Delta f}^{M-1}\left(\left(I_{ca}+I_{cm}\right)M^2-I_{ca}\right)+M^2N_{FFT}\left(\left(I_{ca}+I_{cm}\right)K-I_{ca}\right)$
	PSO-FFT-MLE	$j_{\max,NDR}I_{NDR}\left(\left(I_{ca}+I_{cm}\right)M^2-I_{ca}\right)+\left(2I_{cm}-I_{ca}\right)M^2N_{FFT}\log_2\left(N_{FFT}\right)$

TABLE II
Simulation Parameters

Parameters	Values
Number of platforms	2 or 3
Carrier frequency	1 GHz
Band width	5 MHz
Average velocity of ADCAR	100 m/s
PRF	1kHz
Number of pulses	8
Formation mode	Co-Directional
Sampling rate	10 MHz
Pulse width T_p	5 us
Transmit waveform	CAN-PCM

distance (RMSE_P) is given as

$$\mathrm{RMSE}_P = \frac{1}{N_m}\sum_{n=1}^{N_m}\left\| e^{j\hat{\psi}_n} - e^{j\psi_{\mathrm{real}}}\right\| \tag{44}$$

where N_m is the number of Monte Carlo simulations. The simulation parameters are described in Table II. The search interval based on the phase parameter is generally set as $[-\pi, \pi]$. The experiments were performed under DR conditions unless noted as NDR conditions. In particular, CAN-PCM [24] is used as the transmit waveform in the simulation experiments and is also widely used in ADCAR.

A. Effect of TCPs Mismatch in RC Stage

In order to describe the relationship between TCPs estimation and RC synthesis performance, we consider the effect of TCPs mismatch on coherent synthesis efficiency in the RC stage. The examples of the effects of Doppler frequency and range cell deviation have been given in Section III. This section mainly investigates the effect of phase parameters on the coherent synthesis efficiency. It is assumed that the deviations of the transmit phase set and receive phase set satisfy a zero-mean normal distribution with variance σ_ψ^t, σ_ψ^r, respectively. We assume SNR $= 10$ dB, and $N_m =$ 500 to obtain the following simulation results.

Fig. 5 shows the relationship between the random phase deviation and the coherent synthesis efficiency. It can be seen that the smaller the variance of the phase deviation, the higher the RC synthesis efficiency. When the phase deviation variance increases to a certain degree, there is a significant decrease in coherent synthesis efficiency, ultimately converging to a specific efficiency. The reason is that the phase values after complex spatial modulation tend to be uniformly distributed [45] as the variance of the phase deviation increases. Comparison of Fig. 5(a) and (b) shows that the ADCAR becomes more sensitive to the phase deviation variance.

Further, we analyzed the effect of single-phase parameter mismatches. Let $\hat{\psi}_m^t$ vary within the interval $[-\pi, \pi]$ and the remaining TCPs are sufficiently accurate. As shown in Fig. 6, it can be observed that as the phase distance increases, the coherent synthesis efficiency gradually decreases, and the corresponding $\mathrm{SNR}_{\mathrm{gain}}$ also decreases subsequently. This effect trend decreases with the increase in the number

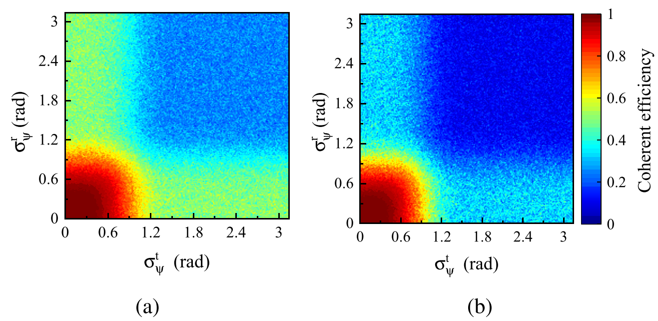

Fig. 5. Relationship between the random phase deviation and the coherent synthesis efficiency. (a) $M = 2$. (b) $M = 3$.

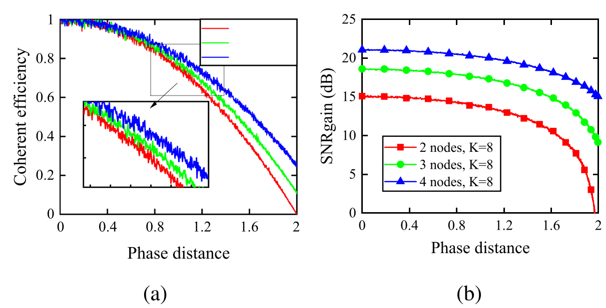

Fig. 6. Effect of single phase deviation on coherent synthesis performance. (a) Single phase versus coherent synthesis efficiency. (b) Single phase versus $\mathrm{SNR}_{\mathrm{gain}}$.

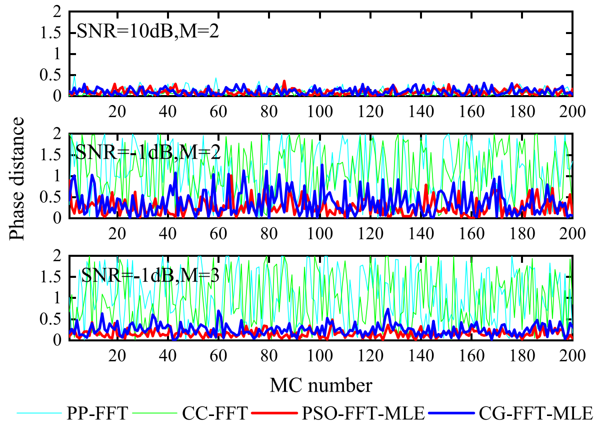

Fig. 7. Floating curve of phase distance for different TCPs estimation methods.

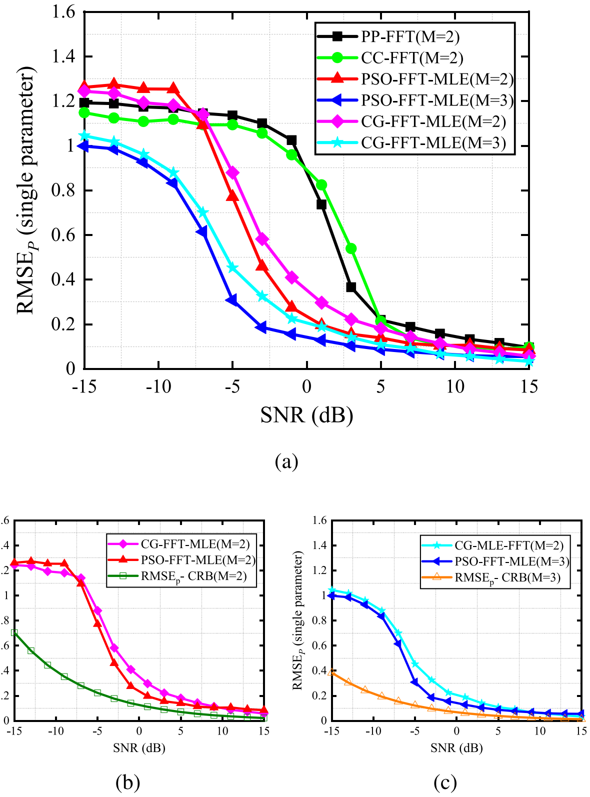

Fig. 8. RMSE$_P$ versus SNR for different TCPs estimation methods.

of nodes. However, the addition of multiple nodes tends to introduce more parameter deviations. Instead, this experiment aims to investigate the effect of a single parameter on the received-coherent performance for different nodes. Therefore, it can be realized that the estimation accuracy of the TCPs determines the performance of the RC synthesis.

B. Performance of RC Synthesis Methods

This section focuses on the performance of the RC synthesis methods with different SNR backgrounds. The ADCAR consists of two or three nodes, with 1st as the reference node. Under the DR condition assumption, the baseline relative to the reference node is 50 m, and $[\theta_{\min}, \theta_{\max}] = [0^\circ, 30^\circ]$. The ADCAR is moving with an average velocity v_0. The velocity deviation of each node from v_0 is not more than 5 m/s. The target is located 50 km from the 1st, and $\theta = 20^\circ$. The relative normalized Doppler frequency of the target is -0.2.

1) Performance of TCPs Estimation: Under DR conditions, the TCPs affecting the coherent synthesis efficiency are mainly phase parameters, i.e., the interchannel differences are mainly caused by phase parameters. We consider the estimation accuracy of the phase parameters with different SNR backgrounds. Let the actual $\Delta\boldsymbol{\psi}$ be randomly generated in each Monte Carlo experiment, and $N_m = 200$ to calculate the phase distance.

The floating curve of phase distance for 200 Monte Carlo experiments is illustrated in Fig. 7. We can see that the proposed methods, PSO-FFT-MLE and CG-FFT-MLE, ensure low-phase distances and floating curves in high and low SNR scenarios, and the proposed methods improve performance further by increasing the number of nodes. The RMSE$_P$ of the TCPs estimation methods with different SNR backgrounds is shown in Fig. 8. It is more evident that the PSO-FFT-MLE and CG-FFT-MLE have better TCPs estimation performance than the PP-FFT and CC-FFT. The above results are because the proposed methods benefit from the joint TCPs estimation for multiple TR pairs. Instead, PP-FFT and CC-FFT do not utilize the target information of all TR pairs. Compared with the parameter estimation accuracy of PSO-FFT-MLE and CG-FFT-MLE, it can be seen that the solution results of CG-FFT-MLE utilizing parameter gradient tend to be more affected by noise. Further, we consider the derivation of the CRB for the phase parameter is given in Appendix B and the proposed method is compared with the CRB, as shown in Fig. 8. In addition, the proposed methods achieve higher accuracy of TCPs estimation in lower SNR scenarios, which is of great significance for the deployment of the TRC stage in ADCAR.

2) Performance of RC Synthesis with DR Condition: Similarly, the RC synthesis results with the DR condition are given under different SNR backgrounds. The results of signal coherent synthesis for different methods in the SNR = 10 dB scenario are shown in Fig. 9(a)–(e), and Fig. 9(f) presents the multipulse accumulation results at the 11th pair. It can be observed that all methods can achieve RC synthesis when the single TR pair can obtain the peak of the target by multipulse accumulation. The SNR improvement of the multichannel RC synthesis is about four times compared with the single TR pair, which aligns with the theoretical. Fig. 10 presents the RC results in the SNR = -6 dB scenario. The PSO-FFT-MLE and CG-FFT-MLE can synthesize the target peak, and the results are close to the optimal parameters. Conversely, PP-FFT and CC-FFT cannot synthesize the target peak due to the low SNR of the single TR pair.

The analysis shows that the proposed methods search in "cell" based on the shared parameters f_d, ρ of all TR pairs with DR conditions while searching for the peak in the phase

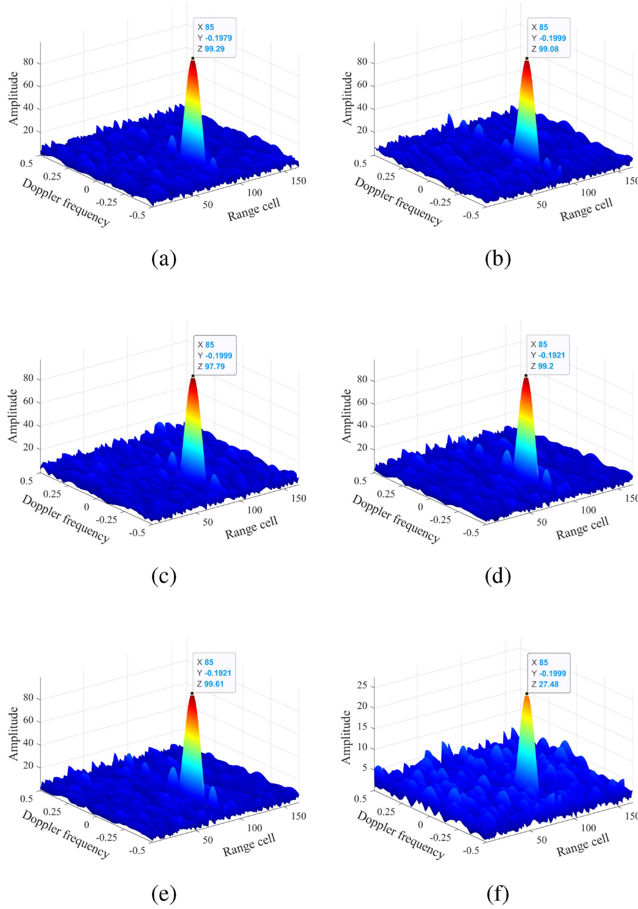

Fig. 9. RC synthesis result for ADCAR with DR condition in the high SNR scenario. $M = 2$, $K = 8$. (a) RC synthesis with the optimal parameters. (b) PSO-FFT-MLE. (c) CG-FFT-MLE. (d) PP-FFT. (e) CC-FFT. (f) 11th with the intra-channel accumulation.

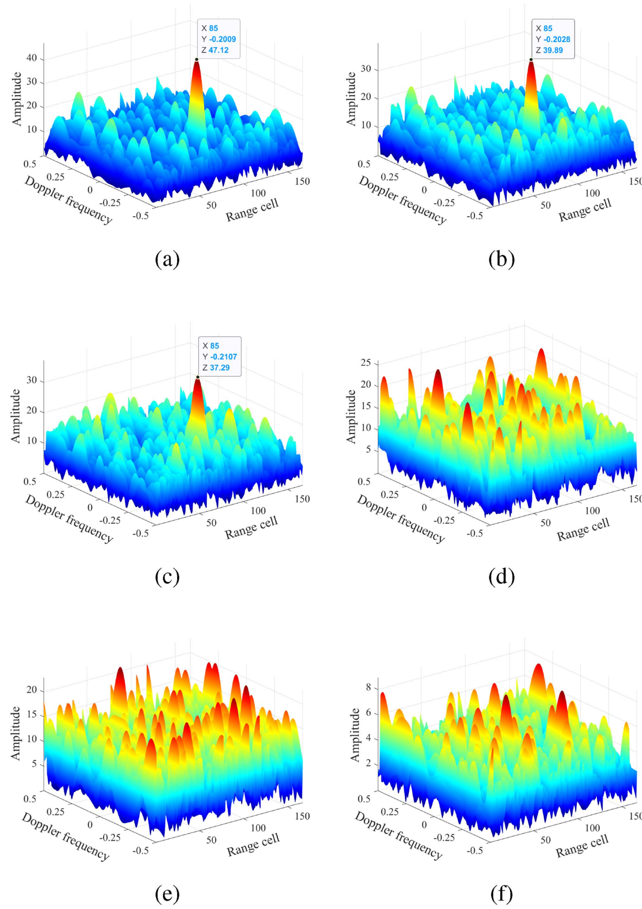

Fig. 10. RC synthesis result for ADCAR with DR condition in the low SNR scenario. $M = 3$, $K = 8$. (a) RC synthesis with the optimal parameters. (b) PSO-FFT-MLE. (c) CG-FFT-MLE. (d) PP-FFT. (e) CC-FFT. (f) 11th with the intra-channel accumulation.

parameters using the PSO and CG methods. It indicates that the proposed methods utilize the target information in all TR pairs to synthesize the target peaks compared to the conventional methods, which is propitious to the following target detection. Combined with the conclusion of the experiments in 1), the proposed methods are corroborated to possess a decent performance of RC synthesis with DR condition.

3) Performance of RC Synthesis with NDR Condition: In practice, some factors (e.g., detection scenarios or synchronization) tend to cause TR pairs to still occur with range cell and Doppler envelope shift after f_d, ρ calibration. Therefore, this section considers the RC synthesis with NDR conditions. For simplicity, it is assumed that there are range cell deviation $\Delta \boldsymbol{d} = \{3, 4\}$ and normalized Doppler deviation $\Delta \boldsymbol{f}_d = \{0.04, 0.06\}$ in the radar echoes of the ADCAR with $M = 2$. In the SNR $= 10$ dB and SNR $= -2$ dB scenarios, the pulse compression results of all TR pairs are represented in Fig. 11(a) and (d), respectively, where the number of pulses for each TR pair is 8, and the total number of pulses is 32. In the high SNR scenario, Fig. 11(b) and (c) show the signal coherent synthesis results of the PSO-FFT-MLE method without considering the NDR condition and considering the NDR condition, respectively. It can be seen that the PSO-FFT-MLE considering $\Delta \boldsymbol{d}$, $\Delta \boldsymbol{f}_d$ focuses on the target signal energy as a prominent peak. In contrast, PSO-FFT-MLE without $\Delta \boldsymbol{d}$, $\Delta \boldsymbol{f}_d$ cannot obtain a high coherent synthesis efficiency even in high SNR scenarios. Similarly, in the low SNR scenario, comparing the results of RC synthesis given in Fig. 11(e) with Fig. 11(f), the proposed method considering $\Delta \boldsymbol{d}$, $\Delta \boldsymbol{f}_d$ forms the target peak above the background noise well. Therefore, the RC synthesis strategy with NDR condition should be considered when ADCAR determines that the DR condition cannot be satisfied.

C. RC Synthesis for Multitarget Scenario

In practice, there might be multiple targets throughout the detection airspace. We consider three moving targets to verify the RC synthesis ability of the proposed methods for multiple targets. We divide the entire detection airspace into multiple small "cells" of interest based on range cell and Doppler velocity. It is assumed that there is at most one target in "cell," and then the proposed methods are used for the "cell" of interest to achieve RC synthesis in multitarget scenarios.

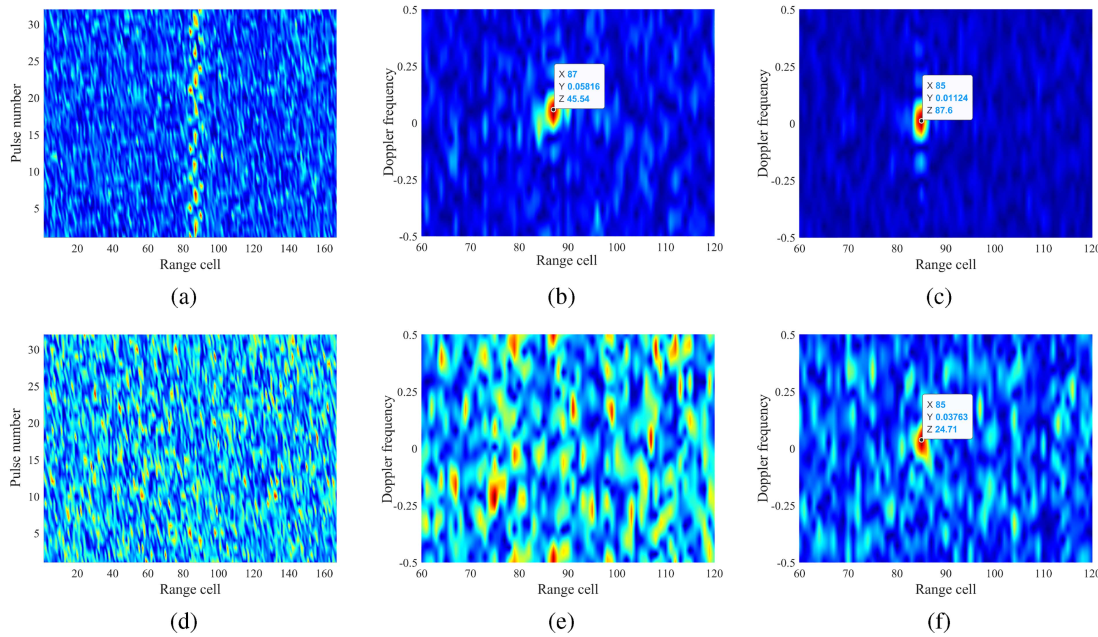

Fig. 11. RC synthesis result for ADCAR with NDR condition. $M = 2$, $K = 8$. (a)–(c) and (d)–(f) are in the SNR = 10 dB and SNR = −2 dB scenarios, respectively. (a) and (d) are the pulse compression results of all TR pairs. (b) and (e) are the signal coherent synthesis results of the PSO-FFT-MLE method without considering the NDR condition. (c) and (f) are the signal coherent synthesis results of the PSO-FFT-MLE method with considering the NDR condition.

TABLE III
Actual and Estimated Parameters of the Target

Target numbers (SNR)	Methods TCPs estimation	Phase values				Normalised Doppler velocity	Range cell	Phase distance
		$\Delta\psi_2^t$	$\Delta\psi_3^t$	$\Delta\psi_2^r$	$\Delta\psi_3^r$			
Target 1 (SNR = -6dB)	Actual values	1.1627	0.8385	1.1779	1.4388	-0.2	85	–
	PSO-FFT-MLE	1.0646	0.9609	0.7710	1.3409	-0.2028	86	0.1481
	CG-FFT-MLE	0.9246	1.1509	1.3754	1.1485	-0.1735	86	0.1751
Target 2 (SNR = -2dB)	Actual values	1.1710	1.0529	0.5698	0.5338	0.3	85	–
	PSO-FFT-MLE	1.3736	1.0853	0.8200	0.7920	0.2937	85	0.1376
	CG-FFT-MLE	1.0577	0.8959	0.7698	0.3567	0.2928	85	0.1098
Target 3 (SNR = 2dB)	Actual values	0.5340	0.1592	-1.0413	1.3125	0.1	55	–
	PSO-FFT-MLE	0.4933	0.1862	-1.2309	1.2708	0.1051	55	0.0666
	CG-FFT-MLE	0.6145	0.2122	-1.3114	1.1242	0.1119	55	0.1141

The simulation parameters for the three targets are given in Table III and $M = 3$. We set the size of "cell" $\{d, \Delta f\}$ to $\{10, 0.2\}$, i.e., the "cell" is formed by ten range cells and the normalized Doppler frequency deviation of 0.2. As shown in Fig. 12(d), with multipulse accumulation in the 11th channel, only the signal energy of target 3 with high SNR is accumulated as a target peak above the noise, and the remaining two targets are still overwhelmed by the noise. The RC synthesis results of the optimal parameters and the proposed methods are given in Fig. 12(a)–(c), respectively. It can be seen that both the optimal parameters and the proposed methods can synthesize all the target peaks. The TCPs estimation results of the corresponding estimation methods are presented in Table III. It can be seen that the proposed methods achieve RC synthesis of multiple targets, and the output TCPs have high accuracy. The average phase distances shown in Table III are in line with the analytical conclusions of the previous experiments and demonstrate the effectiveness of the proposed methods in multitarget scenarios.

D. Detection Performance and Running Time

1) Detection Performance: The probability of false alarm is $P_f = 10^{-3}$ and the detection performance of the signal coherent synthesis with the optimal parameter RC detector, single-G in [46], HCD in [28], and the proposed RC methods are compared via 1000 times Monte Carlo experiments for each SNR. "M2" and "M3" denote the number of platforms corresponding to the detector. Unless otherwise specified, experimental parameters are the same as those in Section V-B. The detection probability of different methods is shown in Fig. 13(a). From this figure, we can see the following.

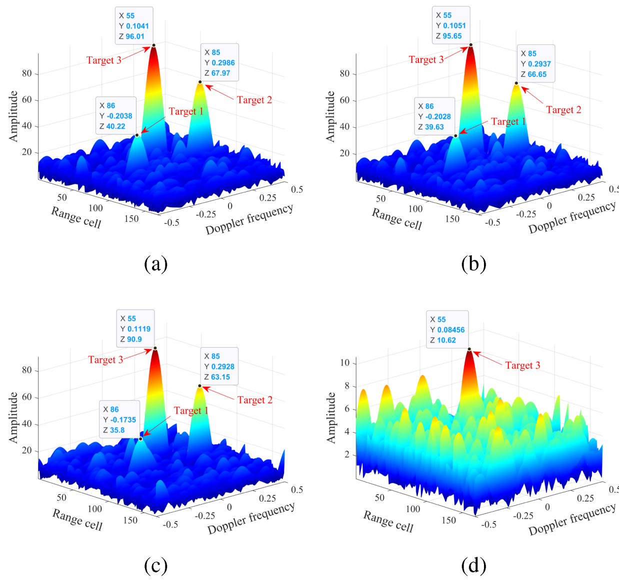

Fig. 12. RC synthesis result for multitarget scenario. $K = 8$. (a) RC synthesis with the optimal parameters. (b) PSO-FFT-MLE. (c) CG-FFT-MLE. (d) 11th with the intra-channel accumulation.

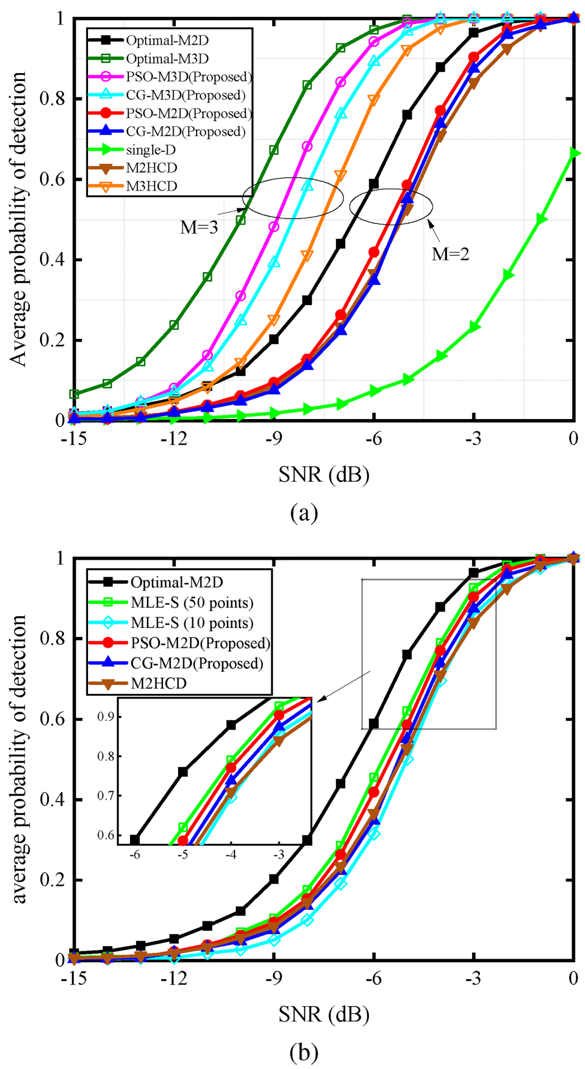

Fig. 13. Detection performance comparison. (a) Different methods. (b) Two fast implementations of RC methods versus MLE-S.

1) Compared with the intrachannel accumulation method with single-G, the SNR improvement of the proposed methods, PSO-FFT-MLE and CG-FFT-MLE, is about 4.4 and 4.2 dB for $M = 2$, and about 7.8 and 7.4 dB for $M = 3$, respectively. This improvement benefits from the signal coherent synthesis of multiple TR pairs in ADCAR.
2) The proposed methods also have decent detection performance compared to the HCD method. This is because the proposed RC synthesis methods match the target's relative phase information between each TR pair. In contrast, the HCD employs partial coherent synthesis within each CPI but noncoherent synthesis across different radars. The above conclusions are more obvious compared to the RC synthetic optimal case.
3) The detection curve shows that the detection ability of the proposed methods is close to the optimal case (the SNR loss is less than 1.2 dB). A comparison of the results in Ssection V-B shows that this is caused by the accuracy loss in TCPs estimation in low SNR scenario. The definition of the optimal detection curve is that we utilize the prior information of the target to compensate for the relative phase deviations and align the envelope shifts among TR pairs, and then obtain the optimal detection result, the previous experiments for optimal parameters were similar. Therefore, the optimal detection curve acquired according to the prior information is used as a benchmark to evaluate the performance of different methods.

Furthermore, the performance of two fast implementations of RC methods is considered. The MLE-S detector in Fig. 13(b) represents the signal coherent synthesis detector based on no closed-form solution of the MLE for the TCPs search, where "50 points" are defined as the number of search points for each phase parameter. As shown in Fig. 13(b), the detection performance of MLE-S with "50 points" is better than that of the MLE-S with "10 points." The proposed methods, PSO-FFT-MLE and CG-FFT-MLE, are in between the two of them. Obviously, the two fast implementations of RC methods demonstrate its advantages by completing the detection faster, as shown in 2) and guaranteeing a lower SNR loss. In addition, the PSO method slightly outperforms the PCG method, as shown by the detection experiments and the results in Section V-B. The reason is that the gradient in the PCG method tends to be corrupted by noise, leading to a bias in the location of the optimal point. Instead, the PSO of the meta-heuristic algorithm reduces the bias by iterating the updates of multiple particles. However, the more considerable computational complexity is given, as shown in 2).

2) Running Time: In this article, the running time is given by the total time taken for the algorithm to run 200 times. The simulation results are obtained on a PC with Matlab R2020a with a 3.8GHz AMD R7 5800X and 16GB of RAM. "d" is defined as range cell search points. The

TABLE IV
Running Time Comparison

Condition	Methods	(M=2) Running Time (s)	(M=3) Running Time (s)
DR	PP-FFT	0.72876	1.6624
	CC-FFT	5.4257	16.456
	PSO-FFT-MLE	103.55	336.63
	CG-FFT-MLE	50.419	110.21
	MLE-S (50 points)	536.63	> 20 min
NDR	PSO-FFT-MLE (d=4)	622.24	> 20 min
	MLE-S (d=4, 50 points)	> 20 min	–

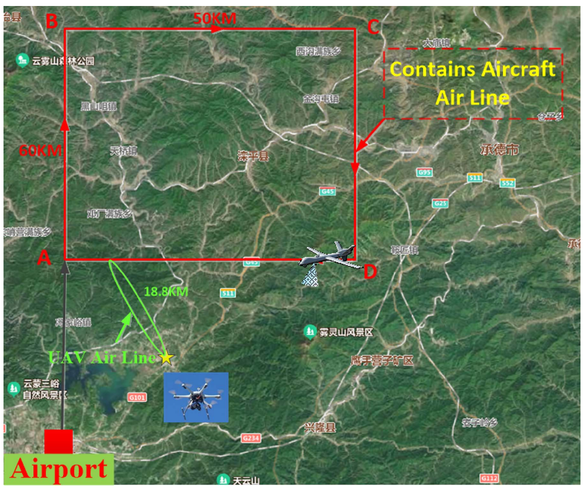

Fig. 14. Schematic of the real experimental scene of ADCAR system.

running time comparison of different methods is shown in Table IV. The multiple TR pairs joint MLE methods have higher computational complexity than the traditional methods. Notably, as the number of nodes increases, the computational complexity of MLE-S with “50 points” increases dramatically. The two fast implementations of RC methods, PSO-FFT-MLE and CG-FFT-MLE, reduce the running time to different degrees with DR and NDR conditions and ensure a low performance loss. In addition, PSO-FFT-MLE utilizing a particle swarm for global optimization leads to longer runtimes compared to CG-FFT-MLE. In summary, the advantages of the proposed methods are demonstrated.

E. Real-Measured Data Results

In this section, an airborne radar system is used to collect radar return data to verify the proposed methods' effectiveness. We have designed an ADCAR system based on CAN-PCM waveforms with the bandwidth of 5 MHz, and the PRF is 3 kHz. ADCAR consists of two distributed radar units with identical performance. A drone target in the detection area moves on a fixed airline. The flight experiment was conducted in a mountainous area with an altitude of about 700–1500 m . A schematic of the ADCAR experimental scenario is illustrated in Fig. 14. The ADCAR first traveled along the airline (red box) with a velocity of about 70 m/s, i.e., the average velocity of the ADCAR. Then,

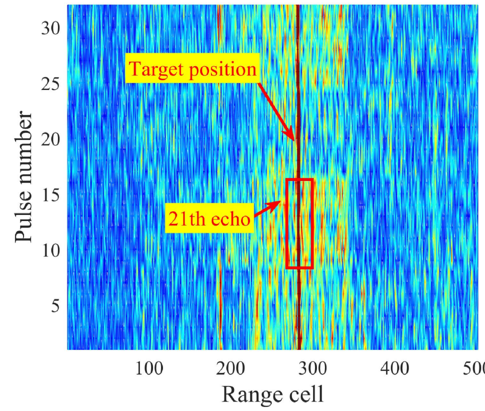

Fig. 15. Pulse compression result for real-measured data after clutter suppression.

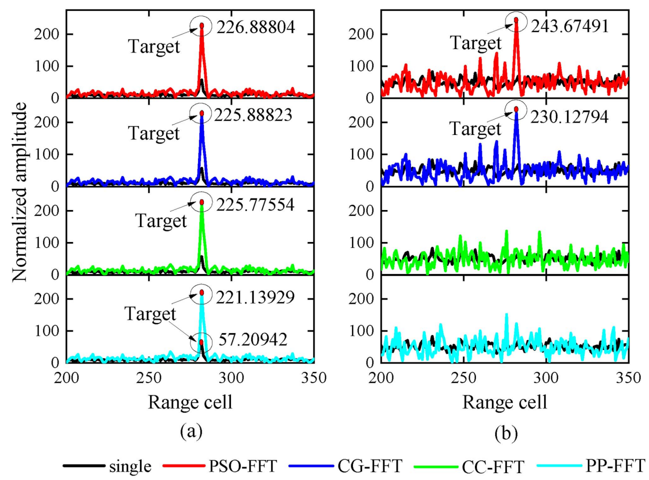

Fig. 16. Real-measured data processing results. (a) RC synthesis result for high-gain target. (b) RC synthesis result for low-gain target.

the drone traveled along the fixed airline with a velocity of 25 m/s.

During the collection of radar data, high-gain and low-gain target radar echoes are collected by adjusting the beam angle of the ADCAR, respectively, and both are processed by clutter suppression. Fig. 15 shows the stacked output signals of four TR pairs for the high-gain target echo. The collected radar echoes are processed by matched filter pulse compression and clutter suppression, and the stack of 8 pulses forms the signal of each TR pair. It can be found that the high-gain target is visible, and its actual detection position and speed are projected to confirm that it is the drone target in the experiment. Fig. 16(a) illustrates the results of RC synthesis for the high-gain target using the different methods, with the corresponding normalized Doppler channel of -0.17. It can be observed that the proposed methods, PSO-FFT-MLE and CG-FFT-MLE, can effectively synthesize the target energy of multiple TR pairs, and the SNR improvement is about 5.44 dB. In addition, since the SNR of the intrachannel is high, the conventional methods, PP-FFT and CC-FFT, also work. The results of RC synthesis for the low-gain target using the different methods are further given in Fig. 16(b). It is demonstrated that the proposed methods also effectively achieve the RC synthesis for the drone target. As a comparison, the PP-FFT and CC-FFT are ineffective.

In this section, the proposed methods, PSO-FFT-MLE and CG-FFT-MLE, are shown to perform superior to the conventional existing methods through various experiments and validation of the real-measured data, which is significantly important for the target detection of ADCAR in the RC stage and the implementation of the TRC stage.

VI. CONCLUSION

This article addressed the signal coherent synthesis of airborne moving target signals problem based on multiple TR pairs for ADCAR in the RC stage. We develop a signal coherent synthesis method based on multiple TR pairs joint TCPs estimation to focus the target's energy distributed among multichannel echoes. Our main contributions include the following: A parameter space division using the DR condition for TCPs is proposed to divide the probe space of interest, and specific arithmetic examples in the ideal case are given; a multiple TR pairs joint TCPs estimation method is proposed for ADCAR in the RC stage, and the corresponding algorithmic strategy are presented combined with the DR and NDR condition; two fast implementations of RC methods (PSO-FFT-MLE and CG-FFT-MLE) are proposed for the signal coherent synthesis and the TCPs output. Results indicated that the performance of the proposed methods for signal coherent synthesis, TCPs estimation, and target detection under different SNR backgrounds is significantly improved compared with the traditional algorithms, and the performance of the proposed methods becomes better with the increase of radar node numbers. Furthermore, deploying the proposed method using the "cell" strategy demonstrates its effectiveness in multiple target scenarios. The effectiveness of the proposed algorithm was validated by real-measured data.

APPENDIX A
DERIVATION OF GRADIENT FOR $\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}}$

In this appendix, we will derive the derivative of $f(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y})$ with respect to $\boldsymbol{\eta}_n^{\mathrm{DR}}$. First, recall that the function (40) is given by

$$f\left(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y}\right) = \left(\mathbf{s}_d{}^H(\mathbf{I}_K \otimes \mathbf{s}_{tr})^H\mathbf{y}\right)^H \left(\mathbf{s}_d{}^H(\mathbf{I}_K \otimes \mathbf{s}_{tr})^H\mathbf{y}\right). \tag{45}$$

Then, the derivative of $f(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y})$ with respect to ψ_m^t and ψ_h^r becomes

$$\frac{\partial f\left(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y}\right)}{\partial\psi_m^t} = \left(\mathbf{s}_d{}^H\left(\mathbf{I}_K \otimes \frac{\partial\mathbf{s}_{tr}}{\partial\psi_m^t}\right)^H\mathbf{y}\right)^H \left(\mathbf{s}_d{}^H(\mathbf{I}_K \otimes \mathbf{s}_{tr})^H\mathbf{y}\right) + \left(\mathbf{s}_d{}^H(\mathbf{I}_K \otimes \mathbf{s}_{tr})^H\mathbf{y}\right)^H \left(\mathbf{s}_d{}^H\left(\mathbf{I}_K \otimes \frac{\partial\mathbf{s}_{tr}}{\partial\psi_m^t}\right)^H\mathbf{y}\right) \tag{46}$$

$$\frac{\partial f\left(\boldsymbol{\eta}^{\mathrm{DR}}|\mathbf{y}\right)}{\partial\psi_h^r} = \left(\mathbf{s}_d{}^H\left(\mathbf{I}_K \otimes \frac{\partial\mathbf{s}_{tr}}{\partial\psi_h^r}\right)^H\mathbf{y}\right)^H \left(\mathbf{s}_d{}^H(\mathbf{I}_K \otimes \mathbf{s}_{tr})^H\mathbf{y}\right) + \left(\mathbf{s}_d{}^H(\mathbf{I}_K \otimes \mathbf{s}_{tr})^H\mathbf{y}\right)^H \left(\mathbf{s}_d{}^H\left(\mathbf{I}_K \otimes \frac{\partial\mathbf{s}_{tr}}{\partial\psi_h^r}\right)^H\mathbf{y}\right) \tag{47}$$

where,

$$\begin{aligned}\frac{\partial\mathbf{s}_{tr}}{\partial\psi_m^t} &= \frac{\partial\mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right)}{\partial\psi_m^t} \otimes \mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right)\\ &= j\left[0,\ldots,e^{j\Delta\psi_m^t},\ldots,0\right]^T \otimes \mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right)\end{aligned} \tag{48}$$

$$\begin{aligned}\frac{\partial\mathbf{s}_{tr}}{\partial\psi_h^r} &= \mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) \otimes \frac{\partial\mathbf{a}_R\left(\Delta\boldsymbol{\psi}^r\right)}{\partial\psi_h^r}\\ &= j\mathbf{a}_T\left(\Delta\boldsymbol{\psi}^t\right) \otimes \left[0,\ldots,e^{j\Delta\psi_h^r},\ldots,0\right]^T.\end{aligned} \tag{49}$$

The above equation is brought into the unknown parameter partial derivatives, and the gradient $\nabla\hat{\boldsymbol{\eta}}_n^{\mathrm{DR}}$ of the corresponding data point can be calculated.

APPENDIX B
DERIVATION OF CRB FOR RMSE$_P$

The lower bound of the RMSE$_P$ for the unbiased estimator is provided by the CRB. Let $\mathbf{p} = e^{j\hat{\boldsymbol{\eta}}^{\mathrm{DR}}}$, and the following equation based on (44) is given by

$$E\left\{(\hat{\mathbf{p}}_{ML} - \mathbf{p})(\hat{\mathbf{p}}_{ML} - \mathbf{p})^H\right\} \geq \mathbf{J}^{-1}(\mathbf{p}). \tag{50}$$

According to the vector CRB transform, we have

$$\mathbf{J}^{-1}(\mathbf{p}) = (\Delta\boldsymbol{\vartheta})\,\mathbf{J}^{-1}(\boldsymbol{\vartheta})\,(\Delta\boldsymbol{\vartheta})^H \tag{51}$$

where $\boldsymbol{\vartheta} = [\psi_{11},\ldots,\psi_{MM}]^T$, $\Delta\boldsymbol{\vartheta}$ is defined as Jacobian matrix, and $[\Delta\boldsymbol{\vartheta}]_{i,j} = \partial \mathrm{p}_i/\partial\vartheta_j$. For (13), the $\mathbf{J}(\boldsymbol{\vartheta})$ is expressed as

$$\begin{aligned}[\mathbf{J}(\boldsymbol{\vartheta})]_{i,j} &= -E\left[\frac{\partial^2 L(\mathbf{y}\,|\boldsymbol{\vartheta})}{\partial\vartheta_i\partial\vartheta_j}\right]\\ &= 2\,\mathrm{Re}\left\{\sigma_t^2\left(\mathbf{u}_j^{\psi}\right)^H\mathbf{R}^{-1}\left(\mathbf{u}_i^{\psi}\right)\right\}\end{aligned} \tag{52}$$

where $\mathbf{u}_i^{\psi} = \frac{\partial\mathbf{u}_i}{\partial\psi_{mh}} = \mathbf{i}_h \otimes [\frac{\partial u_{mh}^0}{\partial\psi_{mh}},\ldots,\frac{\partial u_{mh}^{K-1}}{\partial\psi_{mh}}]^T$, $\mathbf{i}_h$ denotes the vector where the hth element is 1 and others are 0, and $\frac{\partial u_{mh}^k}{\partial\psi_{mh}} = jA_m\xi_{mh}e^{j\psi_{mh}}e^{j2\pi kf_{d,mh}}\chi(\tau_c(l) - \tau_{mh}, 0)$. Based on (8e), (8f), and (44), $\Delta\boldsymbol{\vartheta}$ can be obtained as

$$\Delta\boldsymbol{\vartheta} = \begin{bmatrix}\mathbf{H}_1\\ \mathbf{H}_2\end{bmatrix} \in 2(M-1)\times M^2 \tag{53}$$

where

$$\mathbf{H}_1 = jP^t \odot [\mathbf{H}_{11},\mathbf{H}_{12},\ldots,\mathbf{H}_{1M}] \tag{54a}$$

$$\mathbf{H}_2 = jP^r \odot [\mathbf{H}_{21},\mathbf{H}_{22},\ldots,\mathbf{H}_{2M}] \tag{54b}$$

and $\mathbf{P}^t = (e^{j\Delta\boldsymbol{\psi}^t})^T \otimes \mathbf{1}_{1\times M^2}$, $\mathbf{P}^r = (e^{j\Delta\boldsymbol{\psi}^r})^T \otimes \mathbf{1}_{1\times M^2}$, $\mathbf{H}_{1h} = [-\mathbf{1}_{M-1\times 1}, \mathbf{I}_{M-1}]$, $h = 1,\ldots,M$, and $\mathbf{H}_{21} = -\mathbf{1}_{M-1\times M}$, $\mathbf{H}_{2n} = \begin{bmatrix}\mathbf{0}_{(n-2)\times M}\\ \mathbf{1}_{1\times M}\\ \mathbf{0}_{(M-n)\times M}\end{bmatrix}$, $n = 2,\ldots,M$.

Finally, substituting the above results into (51) and the CRB of the RMSE$_P$ can be obtained by taking the diagonal elements of the matrix.

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